① Lecture 11.02

Memory Architecture

i → Shared Memory Arch.

ii → Distributed " "

iii → Hybrid " "

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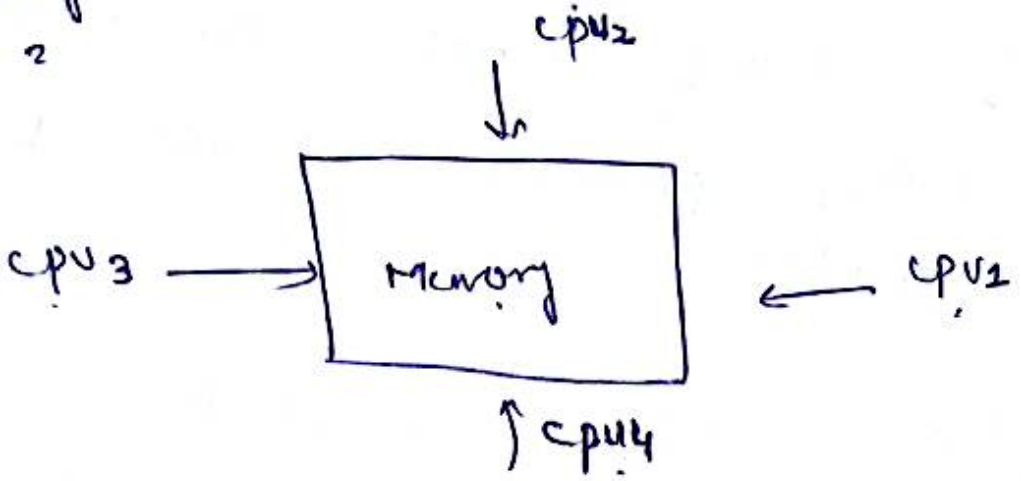
① Shared Memory.

↓

Common memory for all processors

e.g



Keypoints.

i → Memory Access time will remain same for all processors.

ii → If cpu1 is Accessing Remaining all will be in idle state

(2)

Shared Memory

→ Memory Access time is uniform for all

→ Uniform memory Access (UMA)

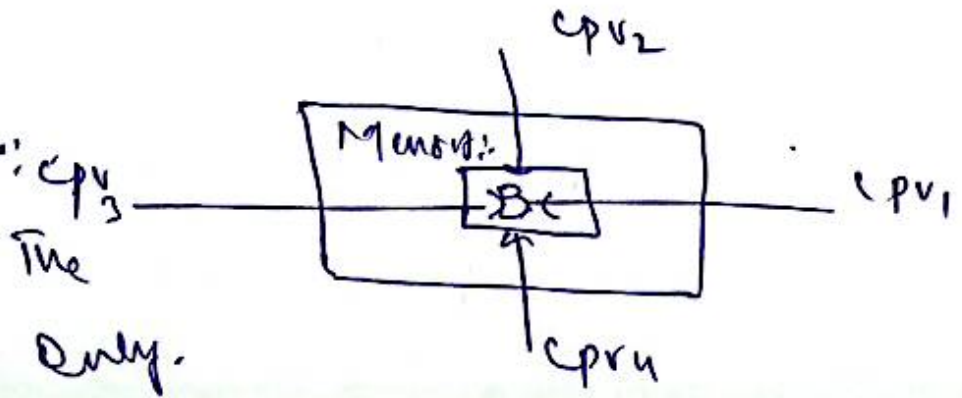
→ issue,

If CPU1 updates the memory, Other processor will not be notified.

Solution: CPU1 → update.
→ shared other with CPU's.

procedure: Cache coherence UMA.

Concept:



i→ update the Buffer Only.

③ **Issue** :- e.g

$CPU_1 \rightarrow$ update
↳ Buffer.

① $CPU_2 \rightarrow$ Same data access $\xrightarrow{\text{find}}$ Buffer

Time Calculation 1

② $CPU_1 \rightarrow$ update 'x' in Buffer

$CPU_2 \rightarrow$ required 'y'

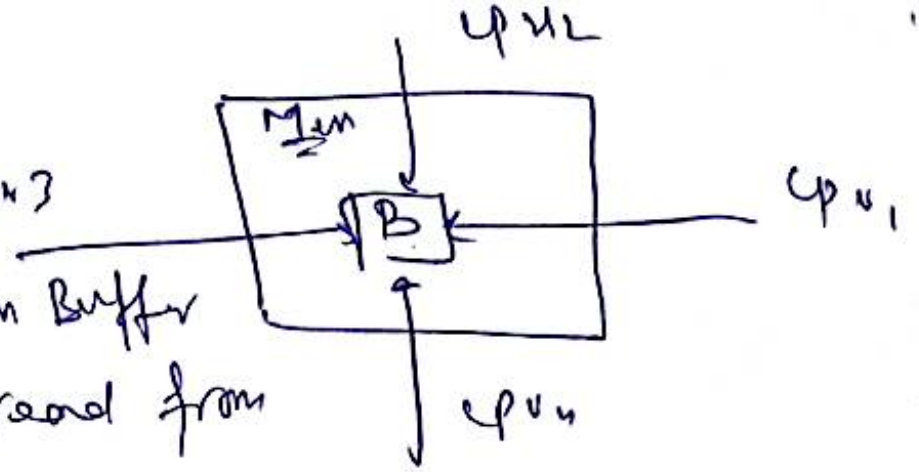
↳ Buffer $\xrightarrow{\text{not found}}$ Memory

Time Calculation 2

Time Cal 1 $\neq$ Time Cal 2

(No UMA)

Refine



→ Write in Buffer but no read from Buffer

(4)

- Now Time for all cpu's will be same

- updated date will be accessed.

Concept :- (C.C. UMA)

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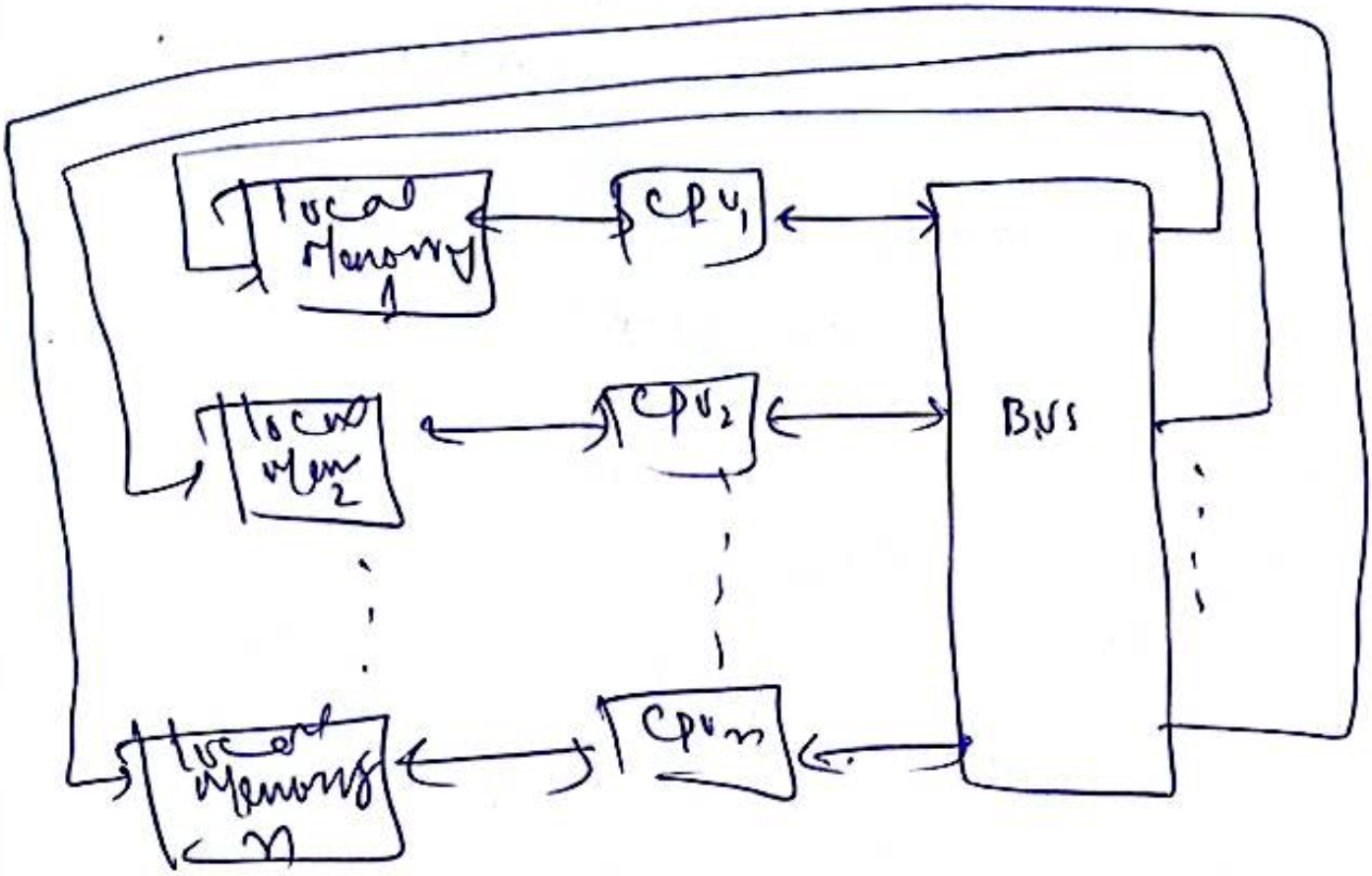
ii → Shared Memory

→ (NUMA) (Non-uniform Memory Access)

↓

~~Different~~ Separate Memory for each processor.



(5) : $cpu_1 \xrightarrow{Access}$ local Memory 1 $\rightarrow$ Time 1

$cpu_2 \xrightarrow{Access}$ local Memory 2 $\rightarrow$ Time 1

But

$cpu_{1,2} \xrightarrow{Access}$ local Memory 2

$\rightarrow$ BUS $\rightarrow$ local Memory 2 $\rightarrow$ Further process.

$\neq$ Time 1

$cpu_1 \xrightarrow{Time 1}$ local Memory 1

$cpu_1 \xrightarrow{Time 2}$ local Memory 2

Time 1 $\neq$ Time 2

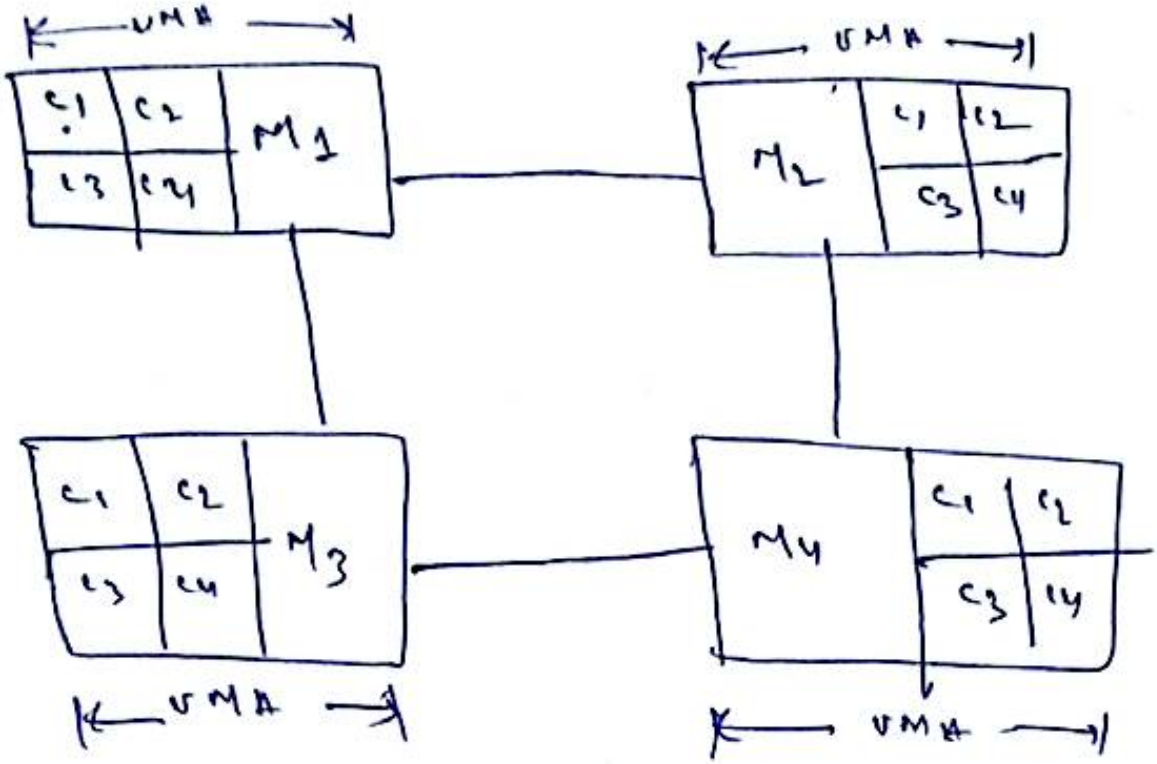
due to change in time.

$\rightarrow$ Non-uniform Memory Access.

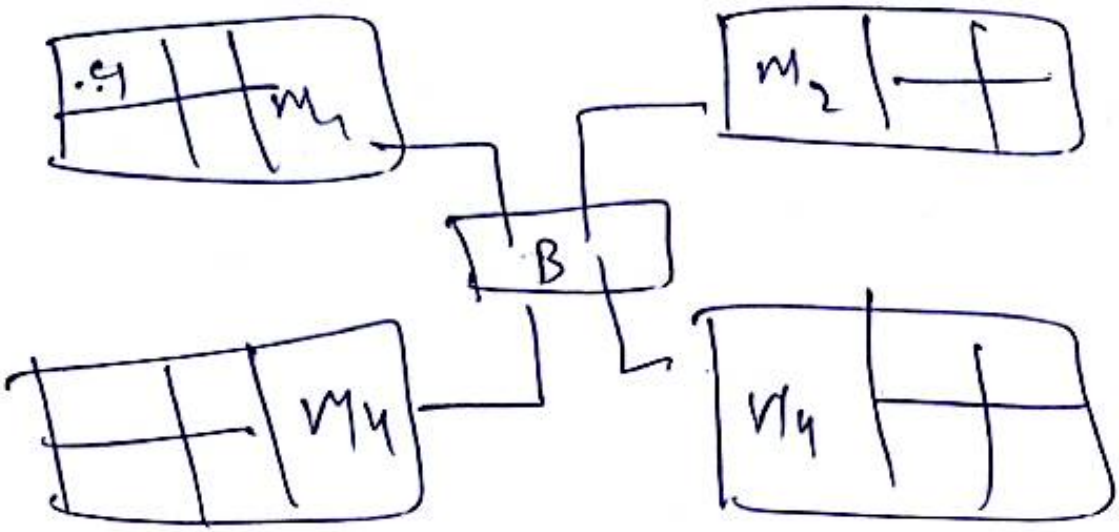
(NUMA)

(16)

Hybrid (UMA + NUMA) :-



$C_1 \xrightarrow{T_1} M_1$

Same $C_1 \xrightarrow{T_4} M_4$

$T_1 \neq T_4$